THE DETERMINANT OF AN ACYCLIC-KERNEL MAP OF RANK ZERO SENDS CANONICAL ELEMENTS TO CANONICAL ELEMENTS

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Allowed standard prerequisites (editorial).

- For a short exact sequence of complexes the determinant of the middle differential corresponds to the tensor product of the determinants of the outer differentials, functorially. invoked at proof lines 37008-37010. Decomposes det(d_K) as det(d_M) tensor det(d_L).

Source reference. Original lemma 0FJC: sources/stacks-more-algebra/pinned-original-source.tex, source lines 36337–36337. Cited at proof lines 37008-37010 for the decomposition det(d_K) = det(d_M) tensor det(d_L); the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let $a^\bullet : K^\bullet \to L^\bullet$ be a map of complexes of R-modules satisfying (1), (2), (3) above. If $L^\bullet$ has rank 0, then $\det(a^\bullet)$ maps the canonical element $\delta(K^\bullet)$ to $\delta(L^\bullet)$.*

Proof. Write $M^i = \mathrm{Ker}(a^i)$. Thus we have a map of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M^{-1} & \longrightarrow & K^{-1} & \longrightarrow & L^{-1} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d_M} & & \downarrow{\scriptstyle d_K} & & \downarrow{\scriptstyle d_L} & & \\ 0 & \longrightarrow & M^0 & \longrightarrow & K^0 & \longrightarrow & L^0 & \longrightarrow & 0 \end{array}$$

By Lemma 0FJC we know that $\det(d_K)$ corresponds to $\det(d_M) \otimes \det(d_L)$ as maps. Unwinding the definitions this gives the required equality. □

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